

Nguyen X. Le

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EDUCATION

Bucknell University

B.S. in Computer Science

Lewisburg, PA

Expected Graduation, May 2027

- **Concentrations:** Software Development and AI & Machine Learning
- **GPA:** 3.60/4.00, Fall 2023, Spring 2024, and Spring 2025 *Dean's List*
- **Related Coursework:** Data Structures & Algorithms, Computer Organization & Programming, Machine Learning, Object-Oriented Programming, Statistics & Applications, Discrete Structures, IOS Development (Codepath)

EXPERIENCE

Brycen

AI Engineer Intern

Hue, Vietnam

Jun. 2025 – Aug. 2025

- Contributed to the development of an **AI-powered company chatbot**, focusing on improving retrieval and response accuracy through research on **Open Deep Search and vector embedding methods (Model2Vec)**.
- **Benchmarked 3+ embedding models** for semantic retrieval to identify best-performing approach.
- Evaluated embedding techniques to support more accurate **information retrieval in chatbot pipelines by 20%**.

Bucknell University

AI/ML Researcher | Python, RAG, Jupyter Notebook, AI

Hybrid, Lewisburg, PA

Feb. 2025 – Present

- Developed a **machine learning-powered lifecycle assessment (LCA) framework** for hydrogen production technologies using OpenLCA and Ecoinvent 3.7.1, reducing manual data processing time by **over 60%** through automation.
- Built and validated **predictive ML models** to estimate environmental impacts (GWP, EP, AP, etc.) of biomass gasification methods.
- Integrated deep research and LangChain-based AI pipeline to extract and structure LCA data from academic literature, enabling **real-time environmental impact prediction** and improving analysis throughput for hydrogen technologies.

Algoverse

AI/ML Researcher | Python, RAG, Jupyter Notebook, AI

Remote, Lewisburg, PA

Jan. 2025 – May 2025

- Developed an **entropy-based dynamic hybrid retrieval algorithm** integrating BM25 and MiniLM-L6-v2 (Hugging Face) to adaptively reweight dense and sparse scores based on per-query uncertainty, achieving **+7.6% improvement** in LLM-as-a-Judge scores on TriviaQA and **got accepted to ICML VecDB 2025**.
- Benchmarked retrieval pipelines on Hugging Face datasets (HotPotQA & TriviaQA), demonstrating **statistically significant performance gains ($p < 0.01$)** over pure dense and static hybrid baselines using iterative multi-round optimization.
- Designed a **lightweight, retriever-agnostic RAG module** with FAISS and Hugging Face Transformers, enabling **~11% faster retrieval** while maintaining or improving semantic groundedness across diverse QA tasks.

PROJECTS

Dungeon Game Simulator | Java, JUnit, Agile Methods, Git

Apr. 2025 - May. 2025

- Collaborated in a team of 4 to build a **2D dungeon-type game using Java** for backend development.
- Followed **agile methodologies** with weekly sprints and developed user stories and user personas.
- Developed strict **JUnit tests** for core non-GUI classes.

Headstarter AI

Software Engineering Fellow | JavaScript, React.js, Firebase, Agile Methods, HTML, CSS

Jul. 2024 – Sep. 2024

- Built and **deployed 4 AI projects in 5 weeks** using React JS, Next.js, Firebase, Clerk, and Vercel, following **agile methodologies** with weekly sprints and incorporated CI/CD practices for iterative deployment.
- **Developed and launched apps** using Next.js, Material-UI for UI, Stripe API, and Firebase for the backend database, improving user interface design and payment processing efficiency.

SKILLS

Programming languages: Java, Python, JavaScript, HTML/CSS, C, MATLAB, Swift (iOS)

Technologies: Google Analytics, Firebase, Material-UI, Pinecone, Clerk, OpenAI, LangChain, RAG, Node.js, React.js

Tools: IntelliJ, Jupyter Notebooks, Git